Department of Electrical and Electronic Engineering Examinations — Confidential

Model Answers and Mark Schemes — First Examiner: J. BARRIA

Paper Code: EE 3.17 — Second Examiner:

Question Number etc. in left margin | Mark allocation in right margin

Q1 a i) $W < 2a+1$ — 2

ii) $U_{\text{selective repeat}} (W < 2a+1) = \dfrac{W(1-P)}{1+2a}$

$U_{\text{go back N}} (W < 2a+1) = \dfrac{W(1-P)}{(1+2a)(1-P+WP)}$

$\dfrac{U_{\text{selective repeat}}}{U_{\text{go back N}}} = 1-P+WP$ — 5

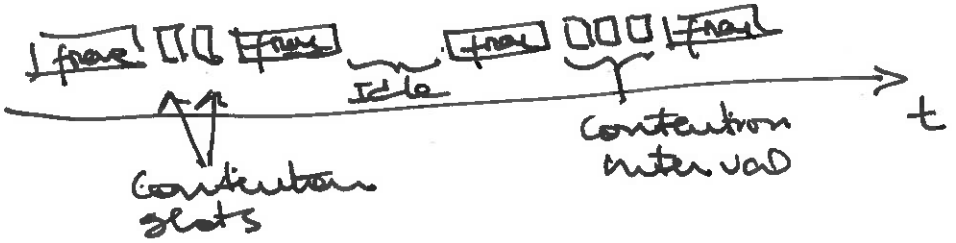
iii) 1 = transmission time (normalized)

a = propagation delay ÷ transmission time (i) & ii))*

W = Window

P = probability a single frame in error — 3

b) i)



(fig 1.1) — 2

ii) given the number of station N and the probability that a station transmit during an available time slot P

A = the prob. that exactly one station transmit

$= \binom{N}{1} P^1 (1-P)^{N-1}$ — 4

iii) probability of j time slots in a row in contention followed by a successful transmission $= A(1-A)^j$

mean number of slots per contention:

$\sum_{j=1}^{\infty} j A(1-A)^j = \dfrac{1-A}{A}$ — 4

Department of Electrical and Electronic Engineering Examinations — Confidential

Model Answers and Mark Schemes — First Examiner: 2019

Paper Code: EE3.12 — Second Examiner:

2/

Question Number etc. in left margin | Mark allocation in right margin

Q2

a)

i) LTE originated from the growth path of 1G, 2G, 3G mobile telecommunication companies and 3GPP (Third generation Partnership Project) which is an organisation consisting of wireless agencies and telecommunication companies. 2

WiMax grew from IEEE standards adopted by data companies with wireless technology e.g IEEE 802.3 and Wifi (802.11) with the aim to mimic the abilities of Wifi wireless internet, but over a mobile phone network. 2

ii)

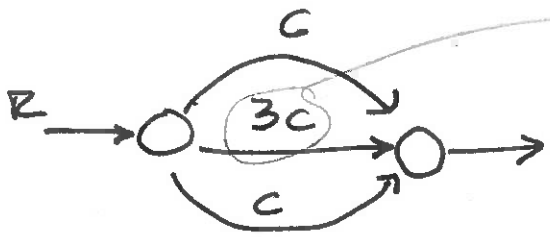
- both are all-IP, packet technology with a packet network core
- both use scalable OFDMA (orthogonal frequency division multiplex access)
- both use (Adaptive modulation and coding (AMC) for link adaptation 3

iii)

- duplex mode
- Access technology
- Spectrum 3

Question Number etc. in left margin | Mark allocation in right margin

26)



$R = x_1 + x_2 + x_3$

$X = x_1 = x_3$

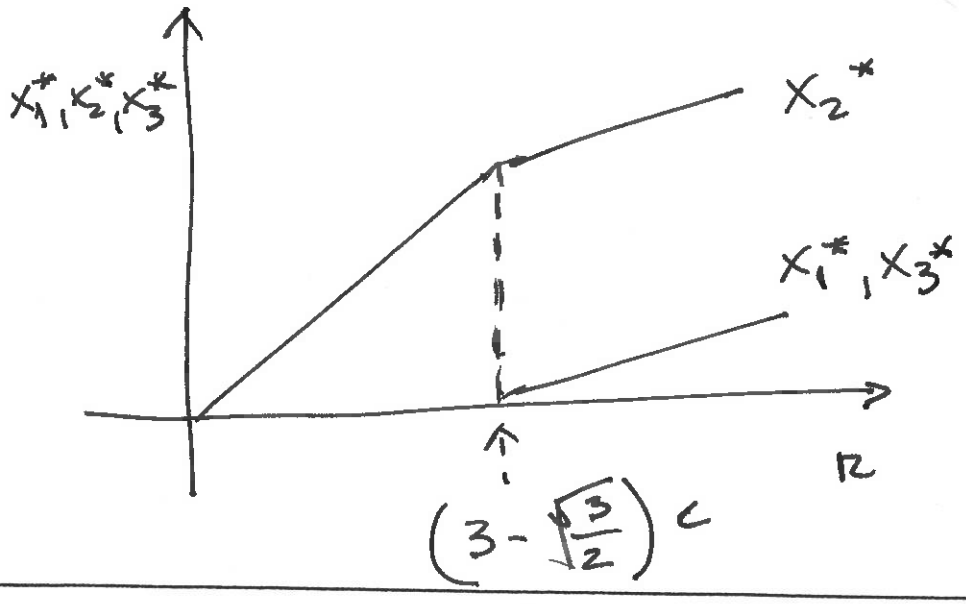
$$D = \frac{x_1}{C - x_1} + \frac{x_2}{3C - x_2} + \frac{x_3}{C - x_3} = \frac{2x}{C - x} + \frac{x_2}{3C - x_2}$$

i) $\frac{\partial D}{\partial x} \rightarrow \frac{2C}{(C-x)^2} = \frac{3C}{(3C - x_2)^2}$ 4

$$2(3C - x_2)^2 = 3(C - x)^2$$

$x = \phi$ as $x_2 = R$

ii) $\sqrt{2}(3C - R) = \sqrt{3}C \rightarrow \left(3 - \sqrt{\frac{3}{2}}\right)C \geq R$ 3



3

1.7752

Question Number etc. in left margin | Mark allocation in right margin

3a i) A host requires three elements to connect to Internet:
- IP address
- subnet mask
- Address to nearby router

each time a user moves these elements need to be reconfigure — 3

ii)
- Host broadcast "DHCP discovery" message (c)
- DHCP server respond with "DHCP offer" message (c)
- Host request IP address "DHCPrequest" message
- DHCP server send address "DHCPAck" message — 3

iii) DHCP:
- IP address
- address of first-hop router for client
- Name and IP address of DNS (Dynamic Name System) server
- Network Mask — 4

b)

i) double cwnd every RTT — 1
if cwnd > Threshold set state = CA.

2) increase cwnd by 1 MSS every RTT — 1

3) Threshold = cwnd/2 — 1
cwnd = Threshold set state = CA

4) Threshold = cwnd/2 — 1
cwnd = 1 set state = SS

5) increase with number — 1
cwnd and Threshold not changed

Question Number etc. in left margin | Mark allocation in right margin

3b ii) Slow start – dynamic cwnd (K)

$$K = \min\left\{k : 2^0 + 2^1 + \dots 2^{k-1} \geq \frac{O}{S}\right\}$$

$$= \min\left\{k : 2^{k-1} \geq \frac{O}{S}\right\}$$

$$= \min\left\{k : k \geq \log_2\left(\frac{O}{S}+1\right)\right\} = \left\lceil \log_2\left(\frac{O}{S}+1\right)\right\rceil$$

K = number of windows that cover the object

$\frac{O}{S}$ = number of segments (S) of the object (O). 2

- after transmitting a window of segments the server could stall
- $\frac{S}{R} + RTT$ = time from when the server receives an ACK for the first segment of a window
- $\left(\frac{S}{R}\right)2^{k-1}$ = transmission time for the kth window
- stall time $= \left[\frac{S}{R} + RTT - 2^{k-1}\left(\frac{S}{R}\right)\right]^+$ 2

(RTT = round trip time)

therefore

$$\text{latency} = 2RTT + \frac{O}{R} + \sum_{k=1}^{K-1}\left[\frac{S}{R} + RTT - 2^{k-1}\frac{S}{R}\right]^+$$

1

Department of Electrical and Electronic Engineering Examinations — Confidential

Model Answers and Mark Schemes — First Examiner:

Paper Code: EE3.17 — Second Examiner: 2019

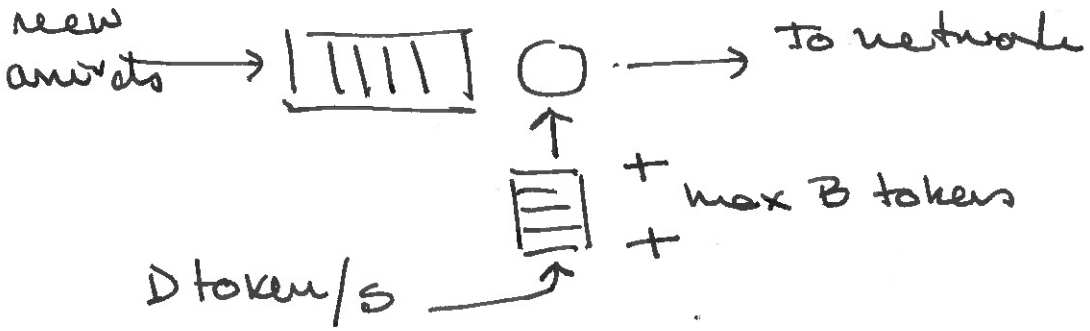
6/

Question Number etc. in left margin | Mark allocation in right margin

Q4 a) The leaky Bucket algorithm can be used to provide a degree of traffic isolation among different classes: Traffic policing. Traffic statistics:
- Average rate of arrival
- peak rate of arrival
- Burst size

3

ii)



3

iii) "Bucket" can hold up to B token
- Token are generated at a rate of D tokens/s when the "Bucket" is full
- Maximum Burst size to network = B
- over an interval of length t: the number of arrivals transmitted to the network is less or equal to $Rt + B$

4

Q4 b) i)
t_i = time sent (timestamp) packet
d_i = delay of ith packet
R_i = time received packet
$v_i = (1-b)v_{i-1} + b|R_i - t_i - d_i|$ $(0<b<1)$
deviation of delay
K = safety constant

4

Question Number etc. in left margin | Mark allocation in right margin

Q4

b)

ii) delay di estimate using an exponential weighted moving average

$$d_i = (1-a)d_{i-1} + a(R_i - t_i) \qquad 0 < a < 1$$

iii) average deviation of delay estimate using an exponential weighted moving average

$$v_i = (1-b)v_{i-1} + b|R_i - t_i - d_i| \qquad 0 < b < 1$$

3

3